01- EQUATION OF STATE FOR IDEAL GAS

PRINCIPLE

The state of a gas is determined by temperature, pressure and amount of substance. In the limiting case of ideal gases, these state variables are related by the ideal gas law. For a change of state under isobaric conditions, this equation becomes Gay-Lussac's first law, while under isochoric conditions it becomes Amonton's law and for isothermal processes it becomes Boyle and Mariotte's law.

TASKS

1. Analyse the validity of the three gas laws for a constant quantity of gas.
2. Calculate the universal gas constant, the coefficient of thermal expansion, and the coefficient of thermal stress using the relationships obtained.

THEORY

The ideal gas law is illustrated in this experiment with the help of three experiments. The aim is to show that this law is a good approximation for the behaviour of many gases under many conditions, despite numerous limitations.

Boyle-Mariotte's law reads: $\boldsymbol{p.V = constant}$ **resp.** $\boldsymbol{p \sim 1/V}$ and states that the pressure of an ideal gas at constant temperature is inversely proportional to the volume. This means that if the pressure of $\boldsymbol{p_1}$ on $\boldsymbol{p_2}$ increases, the volume of $\boldsymbol{V_1}$ on $\boldsymbol{V_2}$ and vice versa:

$$\frac{p_1}{p_2} = \frac{V_1}{V_2} \tag{1}$$

This follows from the formula for the ideal gas law, since in the experiment the temperature T and the amount of substance n remain constant:

$$p.V = n.R.T \quad \text{resp.} \quad p = \frac{n.R.T}{V} \tag{2}$$

This means that the volume of a gas in a gas cylinder under high pressure is less than under normal pressure.

The law of Gay-Lussac is: $\boldsymbol{V/_T\, constant}$ **resp.** $\boldsymbol{V \sim T}$ and states that the volume of an ideal gas at constant pressure is proportional to the temperature. This means that if the temperature of $\boldsymbol{T_1}$ on $\boldsymbol{T_2}$ increases, the volume of $\boldsymbol{V_1}$ on $\boldsymbol{V_2}$ and vice versa:

$$\frac{\boldsymbol{V_1}}{\boldsymbol{V_2}} = \frac{\boldsymbol{T_1}}{\boldsymbol{T_2}} \tag{3}$$

This follows from the formula for the ideal gas law, since in the experiment the pressure and the amount of substance remain constant:

$$\boldsymbol{p.V = n.R.T} \quad \text{resp.} \quad \boldsymbol{V = \frac{n.R.T}{p}} \tag{4}$$

This law is used in pressure vessels, air conditioning systems, balloons and in the analysis of chemical reactions and meteorological conditions.

The law of Amontons reads: $\boldsymbol{p/_T\, constant}$ **resp.** $\boldsymbol{p \sim T}$ and states that the pressure of an ideal gas at constant volume is proportional to the temperature. This means that if the temperature of $\boldsymbol{T_1}$ on $\boldsymbol{T_2}$ increases, the pressure from $\boldsymbol{p_1}$ on $\boldsymbol{p}$ and vice versa:

$$\frac{\boldsymbol{p_1}}{\boldsymbol{p_2}} = \frac{\boldsymbol{T_1}}{\boldsymbol{T_2}} \tag{5}$$

This follows from the formula for the ideal gas law, since in the experiment the volume and the amount of substance remain constant:

$$\boldsymbol{p.V = n.R.T} \quad \text{resp.} \quad \boldsymbol{p = \frac{n.R.T}{V}} \tag{6}$$

In contrast to Gay-Lussac's law, Amonton's law states that at constant volume the *Pressure* of a gas increases when the *Temperature* is increased.

The ideal gas law sets the variables pressure p, temperature T, volume V, and the amount of substance n are related to each other. It is defined as

$$\boldsymbol{p.V = n.R.T} \tag{7}$$

with the Avogadro or ideal gas constant R.

The experimentally obtained diagrams show that there is a linear relationship between the respective external state variables of pressure, temperature and volume analogous to the three gas laws.

The state of a gas is given by the state variables temperature T, pressure p, volume V and the amount of substance n, which interactively determine one another. Thus, the dependence of the volume on the variables temperature and pressure at a constant amount of substance ($n = const.$; $dn = 0$) enclosed amount of gas in the measuring tube) is given by the total differential.

$$dV = \left(\frac{\partial V}{\partial T}\right)_{p,n} dT + \left(\frac{\partial V}{\partial p}\right)_{T,n} dV + \left(\frac{\partial p}{\partial n}\right)_{T,V} dn \tag{8}$$

Analogously, the following is valid for the change of pressure with T and V

$$dp = (\frac{\partial p}{\partial T})_{V,n}\ dT\ +\ (\frac{\partial p}{\partial V})_{T,n} dp + (\frac{\partial V}{\partial n})_{T,V} dn \tag{9}$$

In the evaluation, we will concentrate on the colour-coded terms relevant to the three gas laws.

The relationship (10) is simplified for a certain amount of substance ($n = const.\ ;\ \ dn = 0$ enclosed gas volume in the gas syringe) and an isothermal change of state ($T = const.; dT = 0$) to:

$$dp = (\frac{\partial p}{\partial T})_{V,n}\ dT \tag{10}$$

The partial differential quotient $({}^{\partial p}/_{\partial \boldsymbol{V}})_{T,n} dV$ corresponds geometrically to the slope of a tangent to the function ***V=f(p)*** and thus characterises the interdependence of pressure and volume. The degree of this dependency is characterised by the output pressure p_0 determined. Therefore, one defines the following intensive variables by referring them to V or p or V_0 or p_0 at T_0= 273.15 K:

$$\gamma_0 = \frac{1}{V_0}(\frac{\partial p}{\partial T})_{p,n} \tag{11}$$

$$\beta_0 = \frac{1}{p_0}(\frac{\partial p}{\partial T})_{V,n} \tag{12}$$

$$\chi_0 = -\frac{1}{V_0}(\frac{\partial V}{\partial p})_{T,n} \tag{13}$$

γ_0: coefficient of thermal expansion

β_0: coefficient of thermal tension

χ_0: coefficient of cubic compressibility

For the limiting case of an ideal gas (pressure which is sufficiently small, temperature which is sufficiently high), these relationships can be derived from empirically determined laws, which each describe the correlation between two state variables

For isobaric process control ($p = const.; dp = 0$) eqn. (8) becomes

$$dV = V_0 \gamma_0 dT \tag{14}$$

For $\gamma_0 = const.$, the integration of this differential equation results in

$$V = V_0\left(1 + \gamma_0(T - T_0)\right) = V_0(1 + \gamma_0 \vartheta) \tag{15}$$

ϑ: Temperature on Celcius scale

or

$$\frac{\boldsymbol{V_0}}{\boldsymbol{T_0}} = \frac{\boldsymbol{V}}{\boldsymbol{T}} \tag{16}$$

or

$$V = const.T \tag{17}$$

In accordance with this correlation, which was discovered by Gay-Lussac, the graphic presentation of the volume as a function of the temperature supplies rising straight lines (Fig. 2) with V = 0 for T = 0.

At isochoric conditions (V = const, dV = 0), the following results analogously from (9)

$$dp = p_0 \beta_0 dT \tag{18}$$

For $\beta_0 = const.$, the integration of this differential equation results in

$$p = p_0 \left(1 + \beta_0 (T - T_0)\right) = p_0 (1 + \beta_0 \vartheta) \tag{19}$$

and

$$\frac{\boldsymbol{p_0}}{\boldsymbol{T_0}} = \frac{\boldsymbol{p}}{\boldsymbol{T}} \tag{20}$$

or

$$p = const.T \tag{21}$$

These equations are the expression of Charles' (Amontons') law and describe the linear increase in the pressure with increasing temperature (Fig. 2).

For the isothermal expansion and compression (T = const., dT = 0) of a constant quantity of gas, the following is valid:

$$dV = V_0 \chi_0 dp \tag{22}$$

or

$$V_0 p_0 = V p_0 \tag{23}$$

and

$$V = const.\frac{\mathbf{1}}{\boldsymbol{p}} \tag{24}$$

The graphic presentation of this correlation, which was investi gated by Boyle and Mariotte, results in hyperbolas for V = f(p) (Fig. 3); in contrast, for V = f(1/p) straight lines are obtained (Fig. 4).

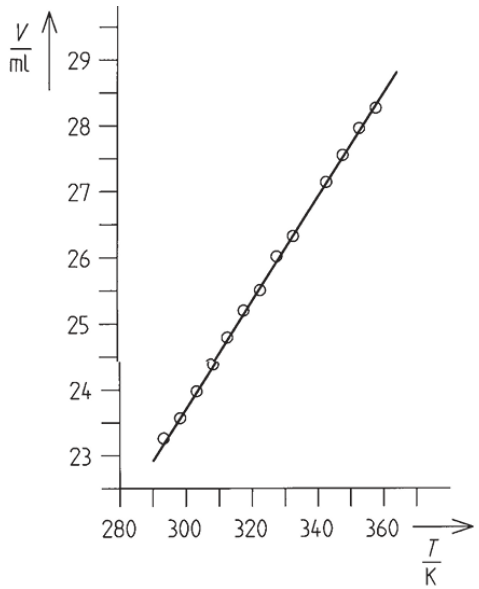


Fig. 2: Dependence ot the volume V on the temperature T at constant pressure (p = 100.6 kPa) and a constant substance quantity (n = 0.9536 mmol).

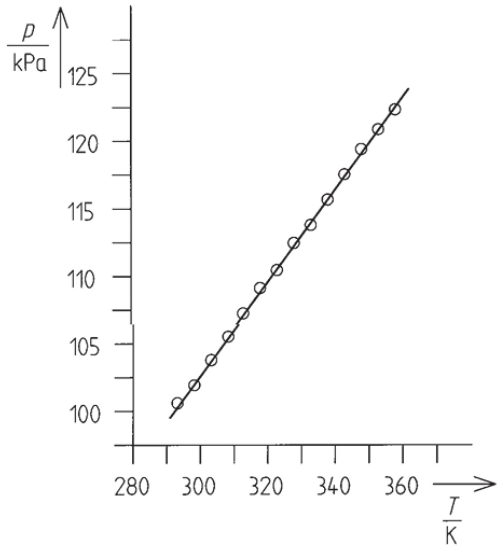


Fig. 3: Pressure p as a function of the temperature T at constant volume (V = 2.326 · 10^{-5} m^3) and a constant substance quantity (n = 0.9536 mmol).

Fig. 4: Correlation between pressure p and volume V for a constant quantity of air (n = 0.9536 mmol) during an isothermic change of state (T = 298.15 K).

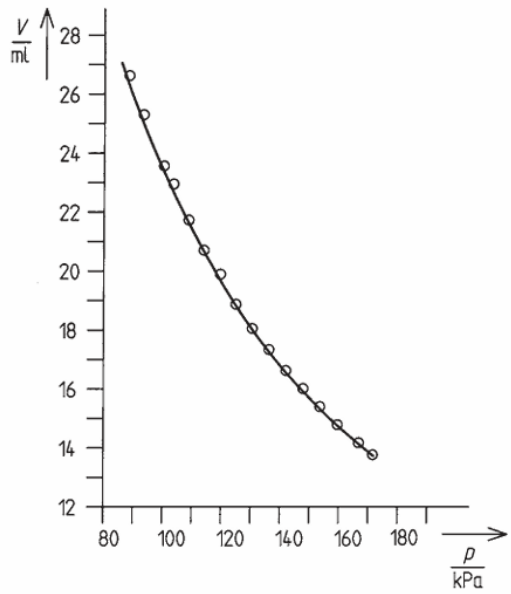
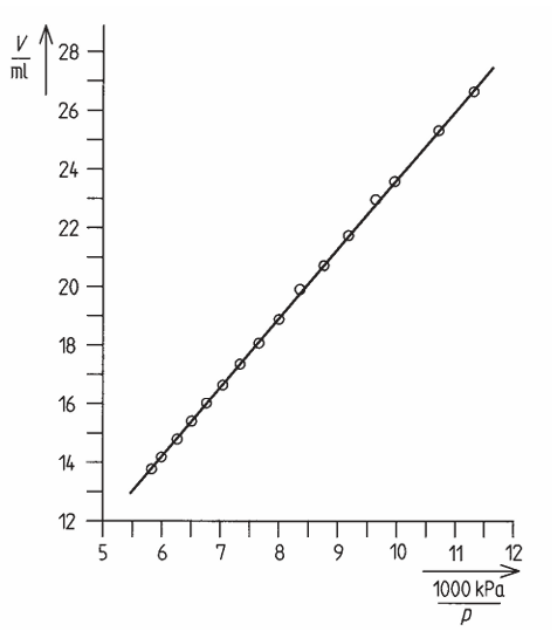


Fig. 5: Volume V as a function of the reciprocal pressure $1/p$ at constant temperature (T= 298.15 K) and a constant substance quantity (n = 0.9536 mmol).



By combining the equations (16) or (17) with (7.1) one direct ly obtains via

$$\frac{p_0 V_O}{T_0} = \frac{p_1 V_1}{T_1} = \frac{pV}{T} \tag{25}$$

the general or thermal equation of state for ideal gases (9) with the universal gas constant R.

$$\boldsymbol{p.V = n.R.T} \tag{26}$$

This equation allows retroactive comprehension of the correlations obtained for equations (17), (21) and (24). Thus from the slopes of these linear relationships.

$$\left(\frac{\partial V}{\partial T}\right)_{p,n} = V_0 \gamma_0 = \frac{nR}{p} \tag{27}$$

$$\left(\frac{\partial p}{\partial T}\right)_{V,n} = V_0 \beta_0 = \frac{nR}{V} \tag{28}$$

$$\left(\frac{\partial V}{\partial p^{-1}}\right)_{T,n} = nRT \tag{29}$$

which can be obtained experimentally, the gas constant R can be calculated if the enclosed quantity of gas n is known. This is equal to the quotient (30) of the volume V and the molar volume V_m.

$$n = \frac{V}{V_m} \tag{30}$$

At T_0 = 273.15 K and p_0 = 101.325 kPa (standard conditions) this amounts to V_0= 0.022414 $m^3 \cdot mol^{-1}$. A volume V measured at p and T must first be reduced to these conditions using Eq. (25).

Moreover, from the equations (27) and (28) the coefficients γ_0 and β_0 can be determined. The required values for the volume V_0 and the pressure p_0 can be obtained by extrapolation to T_0 = 273.15 K on the corresponding graphs or from the corresponding regression equations or calculated using the rela tionships (16) and (20). Therefore, it is possible to calculate the compressibility coefficient x_0 from the total differential (8) using

$$x_0 = \frac{1}{p_0}\frac{y_0}{\beta_0} = \frac{1}{101.325kPa} \cdot \frac{\gamma_0}{\beta_0} \tag{31}$$

The values for γ_0 and β_0 which are now known.

Experiments with a constant quantity of gas (n = 0.9536 mmol) calculated according to the relations (25) and (30) confirm the validity of the investigated gas laws with their resulting linear correlations – which are in accordance with the equations (17), (21) and (24).

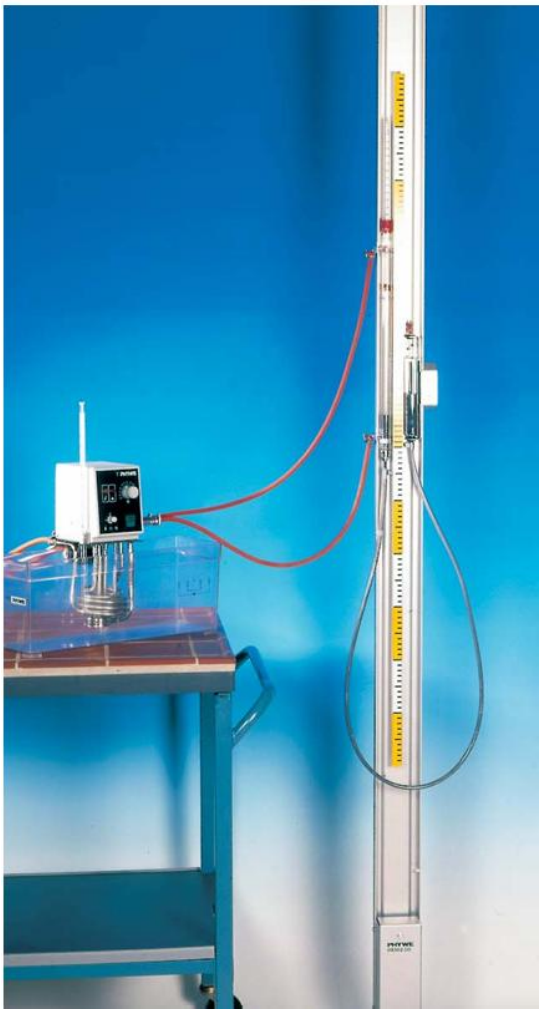
Equipment

Gas law units, thermostat, thermometer, thermostat bath.

Set-up and Procedure

The experiment is set-up according to Fig. 1 and connected with the pump branch unit of the thermostat (Secure the tubing connections with hose clips!). Fill the reservoir on the demonstration device carefully with mercury according to the operating instructions until the measuring tube is filled up to about one-quarter of the gra duated measuring range with mercury (equal levels of mercury in the measuring tube and the levelling container). Fill the bath of the circulating thermostat with distilled or de mineralised water to avoid furring. Connect the cooling coil of the thermostat to the water supply line using rubber tubing

Fig. 1: Experimental set-up: Equation of state of ideal gases



Part I

1. Carefully fill the thermostat bath with water.
2. Set the thermostat to 30°C. Wait until the temperature stabilizes.
3. By moving the gas laws unit, examine the pressure-volume relationship at constant temperature and record it in Table I.
4. Draw the P-V graph. Using the ideal gas equation, calculate the universal gas constant (R) for each measurement. Calculate the mean value (R_{ort}) and the standard deviation (ΔR). Record the values you found in Table I.

Table I

T = sabit= ..... ^{0}C= ……K					
	L(cm)	Δh(cm)	V(ml)	p(kPa)	R(J/mol.K)
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
R= R_{ort} ±ΔR=					

Part II

1. Equalize the mercury columns by moving the gas laws unit at 30°C. Measure the length of the air column L0 corresponding to this situation, calculate the corresponding V0 = constant value, and record it in the tables.
2. Bring the temperature to 40°C. Measure the length of the air column in the measuring tube by setting Δh=0, calculate the corresponding volume value, and record it in Table II. Bring the mercury level in the measuring tube to the L0 position. Read the corresponding εh value, calculate the corresponding pressure value, and record it in Table III.
3. Repeat the same steps, increasing the temperature by 10°C at a time, and take measurements up to 60°C.

4. Using Table II; P=Constant, draw the V-T graph. Calculate the thermal expansion coefficient using equation

(8).

5. Using Table III; Draw the P-T graph where V=Constant. Calculate the thermal stress coefficient using equation **(9).**

Tablo II.

p=sabit=…….....kPa			
t(^{0}C)	L(cm)	V(ml)	T(K)
30			
40			
50			
60			

Tablo III.

V =sabit=… ..... ml			
t(^{0}C)	Δh(cm)	P(kPa)	T(K)
30			
40			
50			
60			

QUESTIONS RELATED TO THE EXPERIMENT

1. The volume of substances in the gaseous state is greater than in their solid and liquid states. Explain why.
2. Are there quantum correlations between the particles that make up an ideal gas? Explain.
3. Which statistic do you think the distribution of an ideal gas follows?
4. What is Avogadro's principle? Research it.
5. Obtain the equation of state for an ideal gas.
6. Derive equation (4).
7. What are the values of the universal gas constant in different units?
8. What is the value of the Boltzmann constant k?
9. What does it mean for a function to be an exact differential equation?
10. Derive equation (17) $\chi 0 = f(\alpha 0, \beta 0)$ from the exact differential equation (6). 11. Why are the expansion and compression coefficients not equal for gases?
11. Which of the physical quantities V, p, T, ε0, ε0, ε0 are intensive and which are extensive?
12. Which of the physical quantities used in this experiment are intensive and which are extensive?
13. What do isothermal expansion and compression mean? Explain.
14. What do isobaric expansion and compression mean? Explain.
15. Does the p-V diagram given in Figure 3 represent the state of real gases?
16. How can we tell that the gas has not liquefied by looking at the diagram given in Figure 3?

17. Is the equation of state given for an ideal gas valid for real gases?
18. Interpret the graph you drew. Are the results as you expected?

REFERENCES

1. PHYWE series of publication, Laboratory Experiments